

TANG LI

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EDUCATION

Liaoning Technical University — M.Eng. in Electrical Engineering (Research Master) Sep 2024 – Jun 2027
Huludao, China Average Score: 92.5/100 (Rank: 1/60)

Relevant coursework: Electric Contact & Arc Theory and Applications (100), Intelligent Electrical Apparatus Principles & Applications (100), Advanced Electromagnetic Field (98), Power Electronics & Motor System Integration (96).

Hefei University — B.Eng. in Automation Sep 2018 – Jul 2022
Hefei, China GPA: 3.3/4

Bachelor's thesis: Design of an Integrated PV-Storage Converter System.

RESEARCH EXPERIENCE

Series Arc Fault Detection — Advisor: Prof. Z. Wang 2025 – Present

- For the published Q1 study identifying series arc faults in frequency-converter-driven three-phase motor circuits *without training on fault samples*, contributed to model development and designed and ran the experiments and comparative benchmarks validating the approach (EAAI).
- Leading an interpretability study (manuscript in preparation): proposed the full concept of *prior-guided time-frequency feature enhancement* for interpretable arc-fault detection under strong background noise, built the model, and wrote the manuscript.
- Developing an *incremental-learning* method for detecting load-fault arcs in three-phase motors under varying noise domains, targeting continual adaptation to new operating conditions (ongoing).

Vision-Based Pantograph Slide Crack Diagnosis — Advisor: Prof. Z. Wang 2024 – 2025

- Carried out the entire study end-to-end — dataset construction, detection-model design, experiments, and manuscript writing — for a vision-based method that detects pantograph-slide cracks to prevent catenary-arcing faults (published in EPSR).

Intelligent Vehicle System — hardware-software co-design 2019 – 2021

- Designed and fabricated the control PCB in Altium Designer (schematic and layout) and hand-soldered the board; optimized the embedded control software for the autonomous vehicle.

PUBLICATIONS & MANUSCRIPTS

***Bold** indicates the applicant; the applicant's specific contribution to each work is described under Research Experience.*

Journal Articles (Published)

- Wang, Z., Li, H., Gao, H., **Tang, L.**, & Guo, F. (2026). An identification model trained without fault samples to identify series arc faults in frequency-converter-controlled three-phase motor circuits. *Engineering Applications of Artificial Intelligence*, 172, 114331. <https://doi.org/10.1016/j.engappai.2026.114331>
- Wang, Z., **Tang, L.**, Liu, S., et al. (2026). Visual diagnosis of pantograph slide cracks for preventing catenary-arcing faults in electric railways. *Electric Power Systems Research*, 261, 113404. <https://doi.org/10.1016/j.epsr.2026.113404>

Conference Papers

- Tang, A., Wang, Z., Gao, H., **Tang, L.**, Guo, F., Wang, L., & Li, C. (2025). Feature extraction method of series arc fault at the back-end of frequency converters. *2025 IEEE 8th International Conference on Automation, Electronics and Electrical Engineering (AUTEEE)*. <https://doi.org/10.1109/AUTEEE67053.2025.11322198>
- Wang, Z., **Tang, L.**, Guo, F., & Lv, X. (2025). A lightweight CNN-GRU model for arc fault detection based on a multimodal feature fusion algorithm. *Proceedings of the 5th International Symposium on Insulation and Discharge Computation of Power Equipment (ISIDCPE)*. https://doi.org/10.1007/978-981-92-0561-5_13
- Hou, Y., Bo, H., Li, S., & **Tang, L.** (2026). Arc detection of series faults in photovoltaic energy storage system. *Proceedings of SPIE*, 14136. <https://doi.org/10.1117/12.3111490>

Manuscript in Preparation

- Wang, Z., **Tang, L.**, et al. Interpretable prior-guided time-frequency feature enhancement for series arc fault detection under strong background noise. (*In preparation.*)

SKILLS

Programming & ML: Python, C/C++ (embedded), PyTorch

Signal Processing & Fault Diagnosis: fault-sample-free anomaly detection, time-frequency analysis, noise-robust modeling

Hardware & Experiment: Altium Designer (PCB design & soldering), STM32/MCU programming, sensor integration & data acquisition, arc-fault test-platform construction

Languages: English — IELTS 6.5 (Overall)